

THE NEW SCHOOL • EUGENE LANG COLLEGE OF LIBERAL
ARTS

Useful Data

Data and Design with Python • LMTH 2080

Monday, August 31, 2026 • 9:00–11:40 AM

Today

1. What makes data useful?
2. Find some data.
3. Who made it, and what's missing?
4. Files and the Unix shell.
5. Keeping track of change with Git.
6. Assignment 0: Find Some Useful Data.

What would you like to know?

From a Problem to Data

problem → question → evidence → data

Think about

Something you would genuinely like to understand, explain, predict, compare, improve, or change.

Data Hunt

Find real data that could help investigate a question.

You are **not** looking for an article that answers the question. You are looking for data that would let you investigate it yourself.

Ask

- ▶ Who collected it?
- ▶ What does one row represent?
- ▶ What variables are available?
- ▶ What could it help answer?
- ▶ What's missing?

Data is made.

What Is Missing?

- ▶ Who or what might be absent from this dataset?
- ▶ What decisions were made before you ever encountered the file?
- ▶ What categories were chosen?
- ▶ What was measured—and what was not?
- ▶ What should someone know before using this data to make a claim or decision?

BREAK

Back at 10:20

Where Am I?

```
pwd  
ls  
cd data  
pwd  
ls
```

Three questions:

1. Where am I?
2. What is here?
3. How do I get where I want to go?

Challenge: Inspect the Data

```
head data/unemployment.csv  
head data/chicago_crimes.csv
```

- ▶ What does one row represent?
- ▶ What variables are available?
- ▶ What question could you investigate?
- ▶ What changes when a dataset has many more columns?

Is head still enough to understand what you have?

Keeping Track of Change

```
git status  
git add <filename>  
git commit -m "Add first notes"  
git log --oneline
```

files change → **Git notices** → **stage** → **commit**

Before You Leave

QUESTION

What is one question you would genuinely like to investigate this semester?

DATA

What data might help you investigate it?

LIMITATION

What might that data fail to capture?

0: Find Some Useful Data

Find **two plausible data sources** for a problem or question that interests you.

For each, identify:

- ▶ where it comes from and who produced it;
- ▶ what one observation represents;
- ▶ important variables;
- ▶ something it could help answer; and
- ▶ an important limitation.

Next meeting: Monday, September 14 — Python as a Tool